

NATIONAL DISASTER INTELLIGENCE AGENCY

WHISPER-LARGE-V3 GEOTECHNICAL INTELLIGENCE REPORT

SITUATIONAL ASSESSMENT: EXECUTIVE · NDMA / ICS-ALIGNED

Report ID:	TEST-DENSE-REPORT-001	Generated:	2026-09-17 14:42
Target Zone:	Kedarnath Valley Scarp	Audit Window:	Real-Time
Coordinates:	Lat 30.7352° · Lon 79.0669°	Elevation:	2015.0 m
Risk Score:	92 / 100	Risk Level:	CRITICAL
Primary Threat:	Landslide / Mass Movement	Trend:	Deteriorating Rapidly
State/Country:	Uttarakhand, India	AI Model:	WHISPER-LARGE-V3

1. WHISPER-LARGE-V3 INTELLIGENCE SYNTHESIS

The following geotechnical assessment was synthesised in real-time by the Whisper-Large-V3 deterministic AI pipeline, integrating live IoT telemetry, tectonic plate modelling, soil mechanics computation, and multi-hazard risk fusion.

CRITICAL geotechnical failure imminent at this location (Lat 30.74, Lon 79.07). Real-time rainfall of 0.0mm has fully saturated the topsoil (98.5%), nullifying slope friction angles. Tectonic stress of 7.80mGal amplifies the hazard.

2. OVERALL RISK SCORECARD

Hazard Vector	Score /100	Risk Level	Probability %	Trigger Mechanism	Evac Zone
LANDSLIDE	92	CRITICAL	97%	Sustained rainfall + antecedent soil moisture	7.4
FLOOD	69	HIGH	72%	Peak hourly rainfall exceeding channel capacity	8.3
EARTHQUAKE	62	MODERATE	64%	Stress accumulation on mapped fault segment	9.3
CYCLONE	15	LOW	15%	Low pressure system intensification	4.5
VOLCANO	20	LOW	20%	Magma chamber overpressure	10.0

Multi-Hazard Severity Profile (0–100 Scale)



3. PER-HAZARD DEEP ANALYSIS & ROOT CAUSE DRIVERS

The following section provides a deterministic root-cause breakdown for each monitored hazard vector, computed from real-time sensor telemetry and Whisper-Large-V3 geophysical synthesis.

■ LANDSLIDE · Landslide

92/100 — CRITICAL

Contributing Factors:

- Rainfall 0.0mm saturating regolith
- Soil saturation 98.5%
- Factor of safety approaching critical threshold
- Steep slope with poor drainage

Trigger Mechanism: Sustained rainfall + antecedent soil moisture

Event Probability: 97% | Estimated Evacuation Zone: 7.4 km²

■ FLOOD · Flash Flood

69/100 — HIGH

Contributing Factors:

- Surface runoff elevated at 0.0mm rainfall
- Impervious surface accumulation
- River bank stress indicators above normal
- Drainage capacity approaching saturation

Trigger Mechanism: Peak hourly rainfall exceeding channel capacity

Event Probability: 72% | Estimated Evacuation Zone: 8.3 km²

■ EARTHQUAKE · Seismic Event

62/100 — MODERATE

Contributing Factors:

- Tectonic stress 7.80mGal above regional baseline
- Fault proximity index elevated
- Lithological amplification factor present
- Recent microseismic cluster detected

Trigger Mechanism: Stress accumulation on mapped fault segment

Event Probability: 64% | Estimated Evacuation Zone: 9.3 km²

■ CYCLONE · Cyclone / High Wind

15/100 — LOW

Contributing Factors:

- Bay of Bengal SST anomaly monitoring active
- Pressure gradient within watch zone
- Wind shear indices within cyclogenesis range
- Track model convergence zone

Trigger Mechanism: Low pressure system intensification

Event Probability: 15% | Estimated Evacuation Zone: 4.5 km²

■ VOLCANO · Volcanic Activity

20/100 — LOW

Contributing Factors:

- Magmatic SO₂ flux within baseline
- Ground deformation GPS: negligible
- Tremor frequency within normal range
- Thermal anomaly monitoring: nominal

Trigger Mechanism: Magma chamber overpressure

Event Probability: 20% | Estimated Evacuation Zone: 10.0 km²

4. VIRTUAL IoT SENSOR TELEMETRY

Real-time physical measurements ingested from the Virtual Sensor Network. Whisper-derived parameters are computed using deterministic geophysical simulation calibrated to the target GPS coordinates.

Sensor Name	Type	Recorded Value	Unit	Health Status	Data Source
Virtual Rainfall Array	Rainfall	0.0	mm	ONLINE	WHISPER-SIM
Virtual Seismic Array	Seismic	7.8	mGal	ONLINE	WHISPER-SIM
Virtual Soil Saturation	Soil Moisture	98.5	%	ONLINE	WHISPER-SIM
Virtual Temperature Sensor	Temperature	20.0	°C	ONLINE	Open-Meteo
Virtual Humidity Sensor	Humidity	65.0	%	ONLINE	Open-Meteo
Virtual Wind Speed Sensor	Wind Speed	10.0	km/h	ONLINE	Open-Meteo

5. GEOTECHNICAL ENGINEERING PARAMETERS

Slope stability analysis computed using Whisper-Large-V3 deterministic geomechanics engine. Parameters are derived from real-time pore pressure, lithological data, and tectonic proximity.

	Value		Value
Slope Angle	24.2°	Soil Cohesion	10.5 kPa
Internal Friction Ang.	14.3°	Factor of Safety	0.6
Stability Status	UNSTABLE	Drainage Condition	Moderately drained
Infiltration Rate	5.6 mm/h	Groundwater Depth	2.1 m
Vegetation Cover	66.8%	Slope Curvature	Convex (dispersion zone)
Lithology	Consolidated sedimentary with clay	Slope Aspect	172.9°

6. 7-DAY PROBABILISTIC RISK FORECAST

Multi-day risk outlook computed by the Whisper-Large-V3 forecast module. Scores represent the expected composite hazard severity index for each 24-hour window.

Day	Date	Risk Score	Risk Level	Primary Threat	Confidence %
D+0	2026-09-18	—	MODERATE	—	95%
D+1	2026-09-19	—	MODERATE	—	89%

Day	Date	Risk Score	Risk Level	Primary Threat	Confidence %
D+2	2026-09-20	—	MODERATE	—	83%
D+3	2026-09-21	—	LOW	—	77%
D+4	2026-09-22	—	LOW	—	71%
D+5	2026-09-23	—	LOW	—	65%
D+6	2026-09-24	—	LOW	—	59%

7. REMEDIAL MEASURES & TACTICAL RESPONSE PLAYBOOK

Response directives generated by the Whisper-Large-V3 SOP engine, aligned with NDMA Incident Command System (ICS) protocols and WHO/UN disaster preparedness guidelines.

7.1 IMMEDIATE RESPONSE (0–6 HOURS)

- Initiate mandatory evacuation
- Deploy rapid-response units to all vulnerable slopes
- Activate Emergency Operations Centre (EOC) — Incident Command System Level 3
- Dispatch NDRF / SDRF rapid-response teams to all vulnerable quadrants
- Issue public broadcast alerts via NDMA Common Alerting Protocol
- Temporarily suspend all construction activities on slopes within 2km radius

7.2 SHORT-TERM ACTIONS (6H–7 DAYS)

- Reinforce retaining walls
- Install real-time slope displacement sensors
- Conduct LiDAR-based slope displacement survey across the monitored sector
- Install additional real-time piezometers at soil saturation hotspots
- Review and upgrade drainage infrastructure — inspect all culverts and spillways
- Audit structural integrity of all load-bearing assets within exposure zone

7.3 LONG-TERM STRUCTURAL MEASURES

- Develop micro-zonation maps at 1:5000 scale for all high-risk wards
- Implement slope stabilisation works (soil nailing, retaining walls, bio-engineering)
- Establish permanent GPS + InSAR ground deformation monitoring network
- Update District Disaster Management Plan (DDMP) with revised hazard exposure data
- Commission detailed Probabilistic Seismic Hazard Analysis (PSHA) study
- Introduce land-use zoning bylaws restricting construction on class-III slopes

7.4 EARLY WARNING TRIGGER THRESHOLDS

- Rainfall threshold trigger: >40mm/hr → auto-alert
- Soil saturation trigger: >80% → pre-evacuation advisory
- Seismic trigger: >Mw 3.5 → structural inspection order
- Slope displacement trigger: >15mm/day → mandatory evacuation

8. POPULATION & INFRASTRUCTURE EXPOSURE

Estimated impact radius computed from Whisper-Large-V3 exposure modelling, incorporating risk score, geographic population density, and spatial hazard extent.

	Estimated Exposure		Estimated Exposure
Roads Km	115.9	Bridges	7
Villages	41	Schools	4
Hospitals	2	Power Lines Km	58.0
Population At Risk	33120	Agricultural Ha	1242.0
Telecom Towers	3	Water Sources	5

9. DATA PROVENANCE & SOURCE REGISTRY

All data streams ingested, validated, quality-checked, and fused by the Whisper-Large-V3 pipeline to produce this report.

Data Source	Type	Latency	Reliability
Open-Meteo API	Meteorological	15-min	99.2%
USGS Earthquake Catalog	Seismic	5-min	99.8%
IMD Rainfall Network	Hydrology	1-hour	96.4%
ISRO Bhuvan LISS-IV	Satellite Imagery	5-day	94.1%
GSI Geological Survey	Lithology	Annual	98.7%
WHISPER-LARGE-V3	AI Geotechnical Synthesis	Real-time	Deterministic